

3D Printing Firm Inflatables with Internal Tethers

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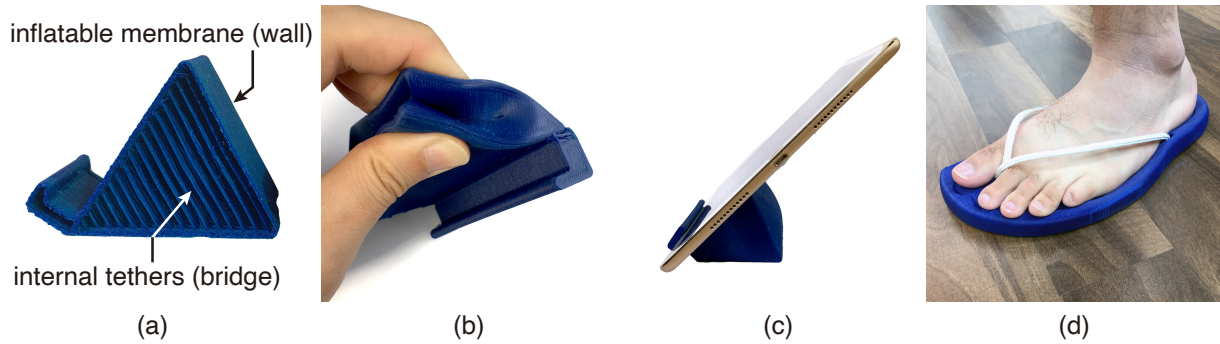


Figure 1: Structure of the 3D-printed inflatables with internal tethers and their application examples: (a) The inflatables consist of flexible membrane and internal tethers so (b) they can be shrunk to reduce the volume. (c) Tablet stand. (d) Flip-flops.

ABSTRACT

This paper presents a technique for 3D printing firm inflatables with consumer-grade fused-deposition modeling (FDM) 3D printers and flexible filaments. By printing bridges inside the inflatable to tie its walls, internal tethers can retain the shape of the surfaces when inflated. This internal structure gives extra stiffness to the inflatables while retaining them lightweight and portable; the inflatables can be squished down to reduce the volume and inflated back to a sturdy state. Compared to conventional drop-stitch fabrics, the length of internal tethers can be easily varied owing to 3D printing, allowing us to fabricate angled surfaces as well as parallel surfaces. We evaluate the physical properties of the 3D-printed inflatables with internal tethers made with diverse printing parameters. Finally, we demonstrate the feasibility of our technique in custom inflatable design with example applications.

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CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

inflatables, internal tethers, FDM, 3D printing

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1 INTRODUCTION AND RELATED WORK

Inflatable objects like boats, tires, or buildings are widely used in our daily lives because they are firm, cost-effective, and easy to deploy [12]. In the field of human-computer interaction (HCI), a number of studies have investigated various techniques for creating inflatable objects and using them as pneumatic user interfaces [4, 17, 21, 26]. The inflatable objects range from simple shapes (e.g., mice [7], buttons [5]) to complex shapes (e.g., teddy bears [25], origami cranes [11]). One of the preferred ways is to stack and weld 2D elastic sheets that can be inflated up to a desired 3D shape [2, 9, 11, 13, 17, 21, 24–26]. *aeroMorph* [11] explored a bending technique

for forming a 3D inflatable from a 2D sheet with various heat-sealing patterns on the sheet. AccordionFab [25] demonstrated techniques for fabricating inflatables from multi-layer sheets by laser cutting and welding the stacked sheets. Printflatables [13] presented a pipeline to fabricate functional and human-scale objects with thermoplastic fabric sheets and heat sealing. Furthermore, design pipeline and fabrication methods for room-scale deployable pneumatic structures have been presented [18, 22].

However, these approaches have two main challenges. First, it is difficult to estimate and design how the fabricated sheet will be transformed into a target shape when inflated. While the bending angle can be controlled by heat sealing methods [11], it is difficult to obtain a planar surface since the sheet inevitably inflates in a spherical shape. The more complex the target shape is, the more difficult it is to calculate the proper heat-sealing patterns on the sheet. Second, the process for fabricating inflatables from the 2D sheets requires a cumbersome process, including cutting the membrane with laser cutters and welding the sheets with heat sealers [6, 11, 13, 20, 24, 25]. While Liquid Printed Pneumatics [16] enables us to directly print complex inflatable structures in a single print, the liquid 3D printer is not accessible to most users. Moreover, the printed inflatables are not firm since the printer utilizes soft silicone materials. Thus, we aim to develop a technique that can 3D-print firm inflatables with commercially-available fused-deposition modeling (FDM) 3D printers.

To realize this goal, we focus on the drop-stitch structure. Drop-stitch structure is one of the promising ways of making firm inflatables by constraining the surfaces' shape with numerous spacer yarns [8]. The structure is made by joining two pieces of woven fabric with a large number of thread yarns. The yarns can apply tension on the inflatable membrane of the object. Consequently, drop-stitch inflatables maintain flat shape even under high pressure, which makes them firm. Moreover, such drop-stitch inflatables can be deflated and folded up for storage similar to the conventional inflatables without adding too much weight. So far, this drop-stitch structure has been used mainly for boats or kayaks that need to be strong enough to withstand the weight of people. In the field of HCI, poimo [10] has been proposed for realizing firm and portable mobility utilizing drop-stitch inflatables. However, the conventional fabrication process, which connects thousands of yarns between fabrics with a knitting or weaving machine, is not feasible to design non-planar shapes, since the length of the spacer yarns cannot be changed at a certain area [23].

In this paper, we present a technique for 3D printing inflatable objects with internally-tethered structure, as shown in Fig. 1. The internal tethers function similar to spacer yarns in drop-stitch structure, which retain the shape of the surface. To 3D print internal tethers inside the inflatable objects, we leverage bridges [3, 19] – horizontal ties between two walls of a print object without any support from below. Since our technique employs consumer-grade fused-deposition modeling (FDM) 3D printers, users can easily fabricate various inflatables by differentiating the length of the tethers (Fig. 1 (a)). For printing materials, flexible rubber-like filaments (e.g., thermoplastic polyurethane (TPU)) were used to allow the object to deflate. The only post-process is inserting a check valve into the air inlet of the inflatable.

Our main contributions are:

- Design workflow for 3D modeling inflatables with custom internal tethers.
- Technical evaluations of the tensile strength of the printed bridges.
- Demonstration of prototype objects and their usage scenarios.

2 DESIGN AND FABRICATION WORKFLOW

Fig. 2 shows the design and fabrication workflow. First, users design the 3D model of the inflatable in a 3D computer-aided design (CAD) software: we used Autodesk Fusion 360¹, a widely-used CAD for designers and makers. After designing the 3D model, a slicing software (we used Cura²) processes the 3D design to generate the G-code (control commands for 3D printers). Next, for printing the object, a common FDM 3D printer capable of printing flexible filaments can be used. We used Creality 3D Ender-5³ with direct drive modification and Artillery Sidewinder X1⁴ with the most common 0.4-mm-diameter nozzle. For the printing materials, we tested NinjaTek NinjaFlex⁵, a thermoplastic polyurethane (TPU) flexible filament of Shore hardness 85A, and DY Air Lastic⁶ flexible filament of Shore hardness 89A. Finally, a check valve was inserted to prevent the compressed air from blowing out. We used Dunlop valves, which are commonly used in bicycle tires, since they were easy to find and low cost. Note that these valves can be replaced with other check valves like air inlets used in swim rings. Users can inflate the printed inflatables through the valve using a common air pump. The printed inflatables can be stored compactly by removing the valve and deflating them.

Fig. 3 illustrates the detailed design flow in the 3D CAD software. First, users design the shape of the inflatable object as a solid body. After that, users create a 2D array of bridges (rectangular bars) to a plane where the internal tethers will be placed (Fig. 3 (a)). The spacing and size of the bridges can be modified by changing the user parameters in Fusion 360. The cross-sectional area of the bridges are designed to be a multiple of the nozzle diameter (e.g., 0.8 mm×0.8 mm). Next, after importing the inflatable object's 3D model, users perform *intersect* command with the object and the bridge array (Fig. 3 (b)), leaving only the bridges inside the model. Then, users execute *shell* command to hollow the model for generating the inflatable membrane (Fig. 3 (c)). The thickness of the membrane can be adjusted at this point: we used 0.8 mm (two perimeters) for most cases except for the top layers. As the bottom part of the top layers is printed in air without support, the number of layers needs to be increased to make them airtight, so the thickness of the top layers is set to 1.5 mm. In our prototypes, if the thickness of the membrane was only one or two perimeters, the nozzle caused scratches on the outer perimeter when traveling from one wall to another, causing microscopic scratches that cause air leakage as shown in Fig. 4. To prevent this, users can optionally add extra perimeters to layers where the bridges exist (Fig. 3 (d)). Finally,

¹<https://www.autodesk.com/products/fusion-360/overview>

²<https://ultimaker.com/software/ultimaker-cura>

³<https://www.creality.com/goods-detail/ender-5-3d-printer>

⁴<https://artillery3d.com/collections/3d-printer-371>

⁵<https://ninjatek.com/ninjabflex/>

⁶https://dyairkorea.com/product/detail.html?product_no=41

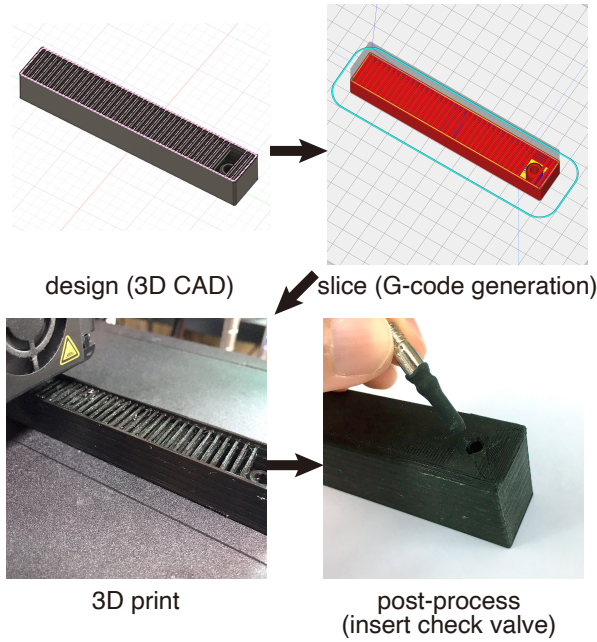


Figure 2: Design and Fabrication Workflow.

users place the pre-designed Dunlop valve insert from the library, then seal the object to complete the inflatable membrane (Fig. 3 (e)).

In the next step, slicer software processes the 3D model to generate G-code. To create airtight prints, we referred to the printing parameters by Yap *et al.* [27], which created soft pneumatic actuators with FDM printing. The important parameters in our settings were as follows: the printing temperature was set high (240°C), and the layer height was set low (0.1 mm) to increase the layer adhesion, and the flow rate was also set high (150%) to fill the gap between layers.

3 TENSILE PROPERTIES OF PRINTED INTERNAL TETHERS

In this section, we explore the tensile properties of the 3D-printed internal tethers to determine their design parameters. The thickness of the tethers determines the tensile strength, but it also has a drawback: the thicker the tethers, the more they can hold the surface shape at higher pressures, while the stiffer the structure becomes, making it hard to be squished. We also assumed that the strength of the tethers is affected by the bridge length because TPU filament is known for its poor bridging characteristics [1]. Thus, we explored the parameters of the tethers considering the following two requirements: 1) if the tethers are soft enough to make the inflatable be squished and 2) the tethers can withstand the desired tension.

The test specimen was composed of two rectangular walls (20 mm × 30 mm) and 9 (= 3 × 3) tethers connected between two surfaces at a 3 mm spacing. The tethers and the walls were printed with NinjaTek NinjaFlex (Shore hardness 85A), and the jigs were

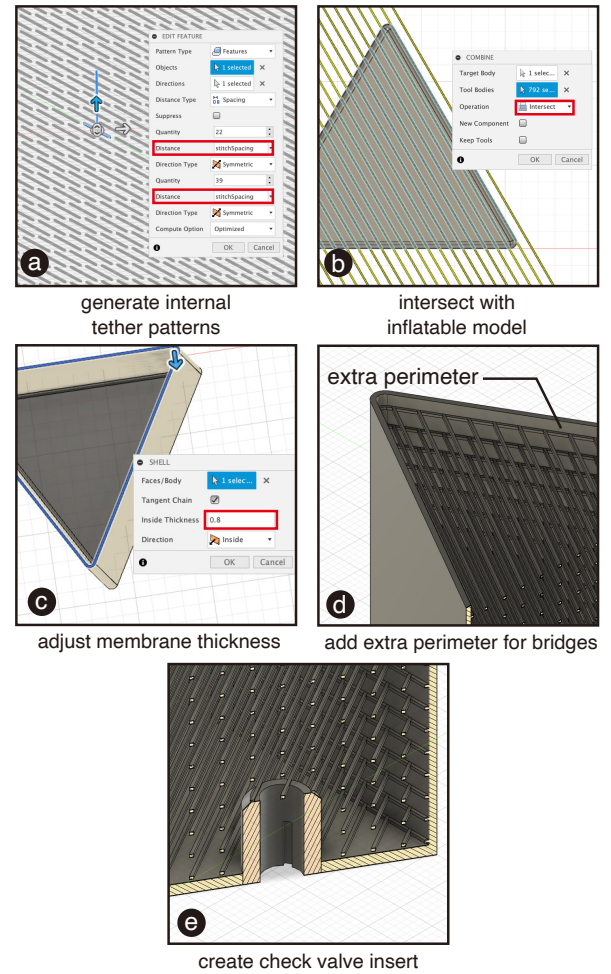


Figure 3: Detailed design process. (a) Generate internal tether patterns with the given parameters. (b) Trim the tethers by intersecting with the model. (c) Adjust the thickness of the inflatable membrane. (d) Add an extra perimeter to the layers where the tethers are placed. (e) Create an air inlet for inserting the check valve.

printed with a regular PLA. Each wall is glued to the jig for attachment to the grip fixture of the tensile testing machine, as shown in Fig. 5 (a). Considering the nozzle diameter of our FDM printer (*i.e.*, 0.4 mm), we modeled the tethers in the CAD software as $0.16 \text{ mm}^2 (= 0.4 \text{ mm} \times 0.4 \text{ mm})$ and $0.64 \text{ mm}^2 (= 0.8 \text{ mm} \times 0.8 \text{ mm})$. After slicing, according to our layer height setting (0.1 mm), this resulted in 4 (1 : bridge × 4 : layers) and 16 (2 × 8) extrusion paths for tethers, respectively. For each thickness, the tether length was ranged from 20 mm to 60 mm with a 20 mm increment. We have not tested tethers sized $1.44 \text{ mm}^2 (= 1.2 \text{ mm} \times 1.2 \text{ mm})$ and over because they were too stiff to be folded. In total, we examined the tensile properties for 6 (= 2 × 3) test specimens.

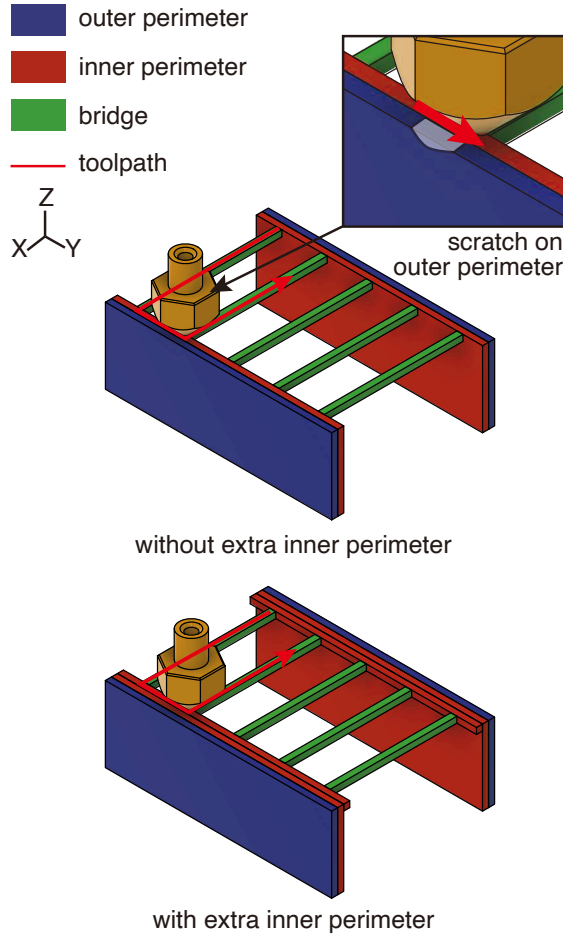


Figure 4: Interference between the nozzle head and the outer perimeter can induce microscopic scratches that cause air leakage. This can be avoided by placing an extra inner perimeter to the layers where the bridges are located.

Using the six specimens, we measured the tensile properties of the specimens with a tensile testing machine (SHIMADZU AUTOGRAPH AG-X⁷), as shown in Fig. 5 (b). The tethers of test specimens were pulled from both sides of the grips while recording the displacement (elongation) and the force (tension). The relationship between the elongation and the tension is depicted in Fig. 5 (c). To discuss the tethers' characteristics, we assume the maximum internal pressure to be 20 psi (1 psi ≈ 6.89 kPa) since our printed inflatables failed to tolerate 30 psi pressure. We used gauge pressure throughout the paper, which is the pressure relative to atmospheric pressure. Each 0.64 mm² tether was elongated approximately less than 20 % under 11.16 N (=1.24 N×9: the total number of extrusions) where 1.24 N refers the force applied to a surface sized 3mm × 3mm (spacing of the tethers) at 20 psi (assumed maximum internal pressure). On the other hand, some of the 0.16 mm² tethers were lengthened more

⁷<https://www.shimadzu.com/an/test/universal/agx-v.html>

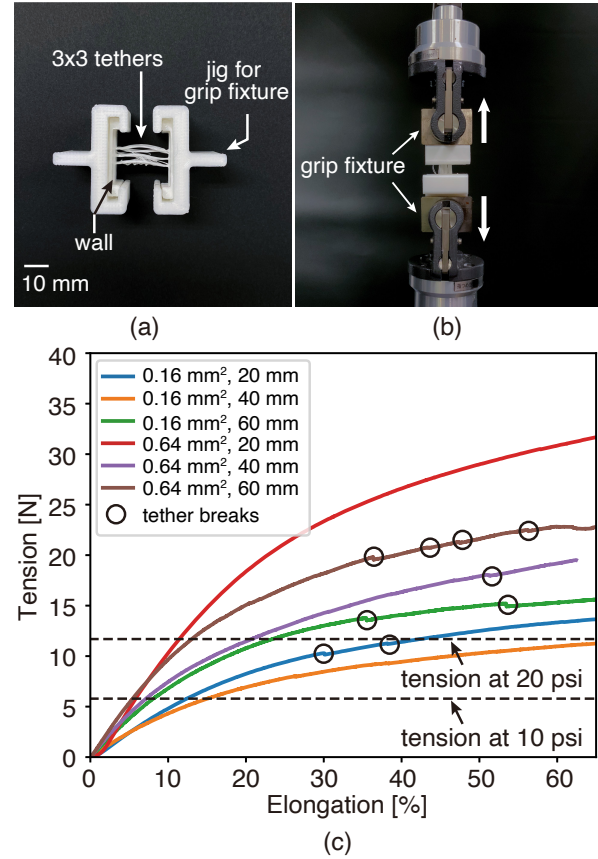


Figure 5: Experimental setup and results of tensile tests. (a) The test specimen. (b) Experiment setup and (c) the relationship between the elongation and the tension.

than 40 % at 11.16 N; this suggests the 0.16 mm² tethers cannot exert sufficient tension to keep the desired surface shape. While at ten psi pressure, the 0.16 mm² tethers were still feasible under 15 % of elongation. Note that the tension drastically decreases at some points due to the breaks of the tethers, which is indicated by the black circles in Fig. 5 (c). Each length of 0.16 mm² and 0.64 mm² tethers withstood the force that corresponds 10 psi and 20 psi, respectively; meanwhile, no clear relationship between the tether length and the tensile property was presented in this experiment. Based on the above experimental results, this study discusses inflatables printed with 0.64 mm² tethers.

4 PRELIMINARY EVALUATION OF OBJECT DEFORMATION

A preliminary evaluation of the deformation was conducted to show the capability of the 3D-printed internally-tethered fibers in retaining the shape of the objects. We prepared two rectangular bar models (width: 20 mm, height: 30 mm, length: 120 mm), as shown in Fig. 6 (a). One had internally tethers arranged in a 3 mm grid spacing, with each tether comprised 2×4 extrusions (modeled as

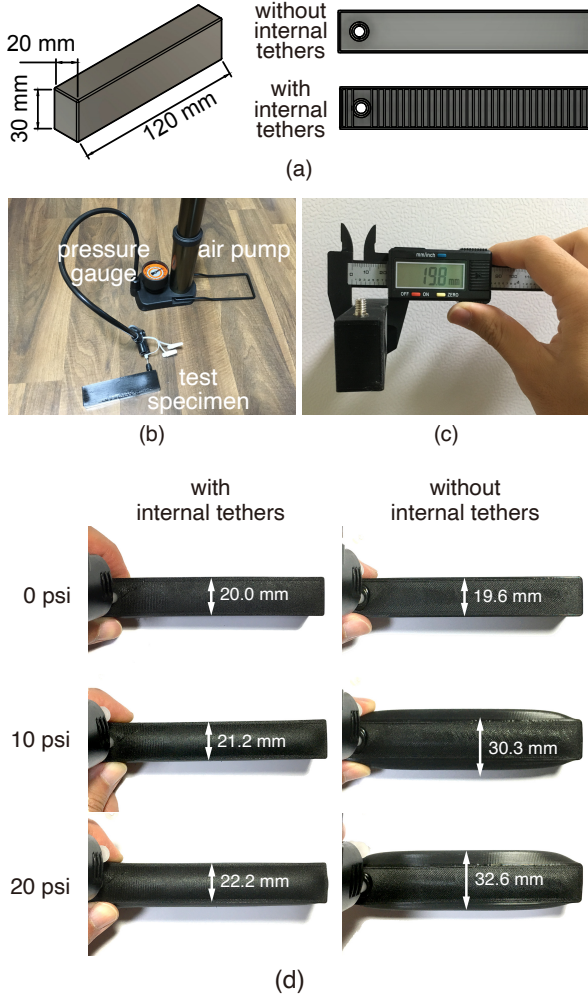


Figure 6: Inflation Experiment. (a) Dimension (left) and a cross-sectional view of the test specimens (right). (b) Experimental setup for inflating the printed objects. (c) Thickness measurement. (d) Thickness comparison of the test specimens with and without internal tethers at different internal pressures.

0.8 mm×0.8 mm), and the other one was hollow. The thickness of the membrane was 0.8 mm, except for the layers where tethers are placed: they had extra internal perimeter for the tethers. The internal pressure was controlled from 0 psi to 20 psi at a 10 psi increment with the air pump shown in Fig. 6 (b). The thickness (width) was measured at the center with a digital caliper (Fig. 6 (c)). During the measurement, the specimen was directly connected to the air pump without the Dunlop valve to accurately control and measure the internal pressure. The results are summarized in Fig. 6 (d). The object with internal tethers showed little increase in thickness as the internal pressure rises, whereas the one without internal tethers expanded drastically even at ten psi.

5 DESIGN EXAMPLES AND APPLICATIONS

To demonstrate the feasibility and the potential of our 3D-printed inflatables, we show two designs and their applications.

5.1 Tablet Stand

To demonstrate that the 3D-printed internal tethers can sustain angled surfaces of the inflatable, we created a simple structure with two angled planes. Fig. 7(a) shows an inflatable tablet stand. With the advantage of 3D printing, the length of the internal tethers can be varied to tension the two flat surfaces when inflated. Also, the stand can be deflated and folded to a compact size for storage: $9 \text{ cm} \times 7 \text{ cm} \times 4.7 \text{ cm} = 296.1 \text{ cm}^3$ when inflated, and $9 \text{ cm} \times 7 \text{ cm} \times 1.9 \text{ cm} = 119.7 \text{ cm}^3$ when folded (40.4 % of inflated state).

5.2 Flip-Flops

To showcase the durability of the 3D-printed inflatables with internal tethers, we designed a pair of inflatable flip-flops, as shown in Fig. 7(c). Compared to conventional flip-flops, our inflatable ones can be squished and folded to make them easy to carry. Taking the advantages of 3D printing, we created a personalized design using the silhouette of the user's feet. The weight of a single flip-flop was about 165 g. The printed flip-flops were able to hold the weight of a person (about 80 kg). The storage volume could be reduced as small as 58.6 % ($29.1 \text{ cm} \times 12 \text{ cm} \times 2 \text{ cm} = 698.4 \text{ cm}^3$ when inflated and $7 \text{ cm} \times 13 \text{ cm} \times 4.5 \text{ cm} = 409.5 \text{ cm}^3$ when folded). It was also easy to adjust the firmness by changing the air pressure, a soft feel at 10 psi, and a firm feel at 20 psi.

6 LIMITATIONS AND FUTURE WORK

6.1 Interactive User Interface for Internal Tether Placement

In the current workflow, the internal tethers are manually designed in 3D modeling software, then processed by slicer software for generating the toolpaths. As the shape of the inflatable gets complex, it becomes difficult to place internal tethers since common CAD software is not suitable for designing complex infill structure. Ideally, it is desirable to implement an interactive user interface for placing the internal tethers into an arbitrary 3D model and generating toolpaths. We plan to build such a user-friendly toolkit and deploy it in the near future.

6.2 Computational Method for Designing Complex Surfaces

While internally-tethered fibers enable to constrain the deformation of the surfaces, there happens a slight deformation due to the elongation of the tethers and the different surface tensions. In the current approach, we only placed the tethers in a certain direction, but we can exploit more design parameters (e.g., thickness, direction, and spacing of the tethers). By leveraging a mass-spring model of inflatable membranes and internal tethers, it is expected to computationally derive the design for complex curved surface structures [14, 15]: for example, to accomplish an organic shape of a computer mouse, the design parameters mentioned above need

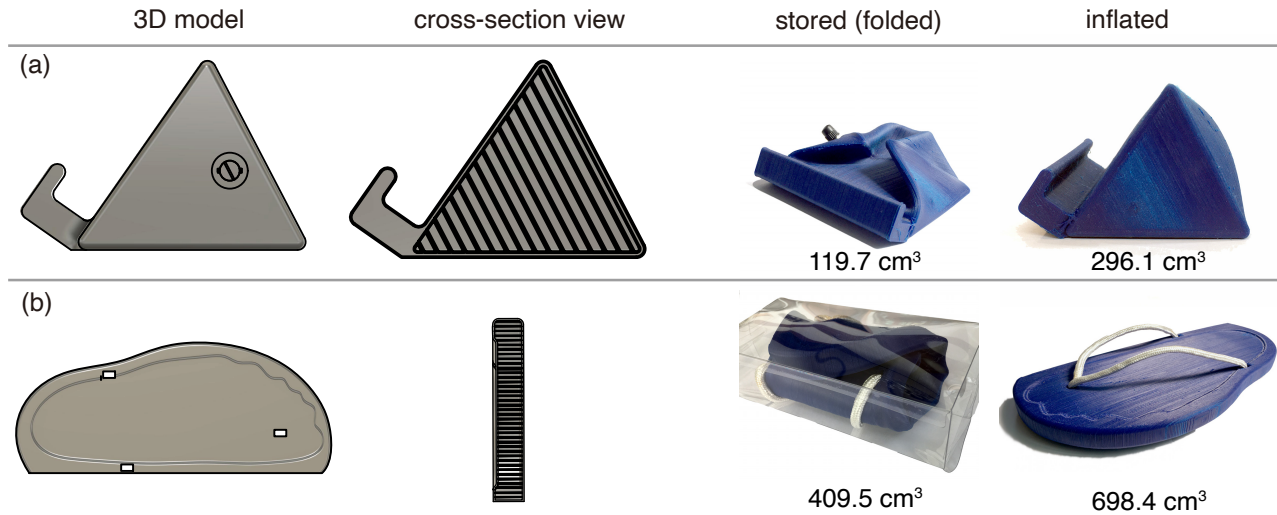


Figure 7: Design examples and printed inflatables. (a) Tablet stand. (b) Flip-flops.

to be calculated and optimized. We intend to explore such computational fabrication methods and integrate them with the interactive user interface.

6.3 Fabrication Method

Although 3D printing offers a substantial opportunity for generating arbitrary shapes, printing inflatables with 3D printers is much slower than rigid prints due to the elastic characteristic of flexible filaments. Furthermore, the endurance of the 3D-printed membrane is not strong enough to bear high internal pressure – in our preliminary evaluation, it only bore up to 30–40 psi of internal pressure. Our future directions will include investigating a computational fabrication method for varying the length of spacer yarns into stronger fabrics, expecting to create stronger inflatables in a short time.

6.4 Motivations for Firm Inflatables

While it is possible to fabricate various complex shapes using cutting and heat-sealing 2D sheets, the firmness of the fabricated inflatables are basically determined by the internal pressure. At an high internal pressure, such inflatables cannot retain their shape because individual air chambers tend to expand in all directions. On the other hand, 3D printing with rigid materials can make strong objects but the volume of them cannot be reduced at ease. By introducing programmable internal tethers in the inflatable manufacturing process, we envision to bridge the design gap between the rigid and soft objects.

7 CONCLUSION

In this paper, we presented a technique for 3D printing firm inflatables with internal tethers. We have described the workflow for designing such inflatables and the fabrication methods. Also, we

explored proper printing parameters for achieving the internal tethers and evaluated the physical properties. Finally, we demonstrated the practicality of our 3D-printed inflatables with two application examples. We hope that our approach encourages researchers and users to introduce internal tethers as a new design element for fabricating inflatables.

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